المعهد الهنـدي للتكنولوجيـــا دلهي
أبـــوظبي

INDIAN INSTITUTE OF TECHNOLOGY DELHI
ABU DHABI

Biomolecules - Proteins

Lecture 12

Prof. Saurabh Raj
Kusuma School of Biological Sciences
IIT Delhi

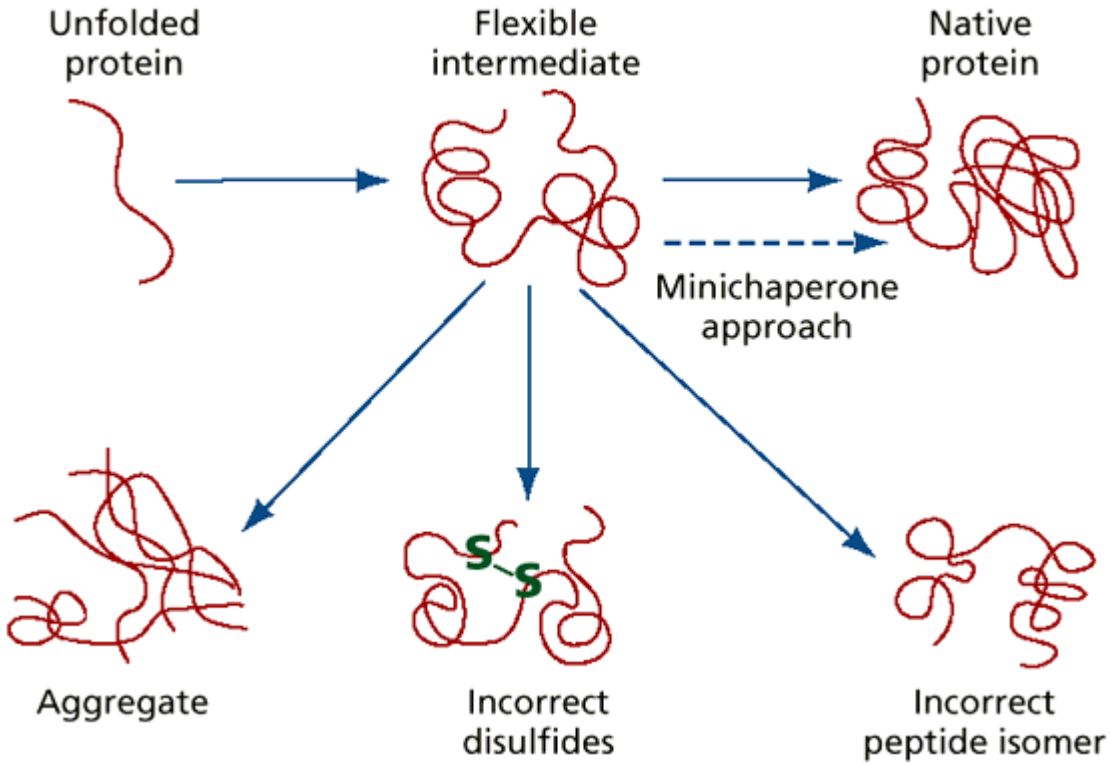
The protein-folding problem

A protein chain has many rotatable bonds.

Each bond can adopt multiple conformations.

The number of possible conformations can become astronomically large.

Yet many proteins fold rapidly and reproducibly.



https://www.nature.com/articles/nbt0299_136

Levinthal's Paradox

Protein folding – the process that results in acquisition of the native structure from a completely or partially unfolded state



Protein folding cannot proceed by purely random search among all possible conformations

For example, 100 AA protein – let's say 2 configurations for each step

It takes at minimum 10^{-12} sec for each step (time scale for bond rotation)

How long to fold?

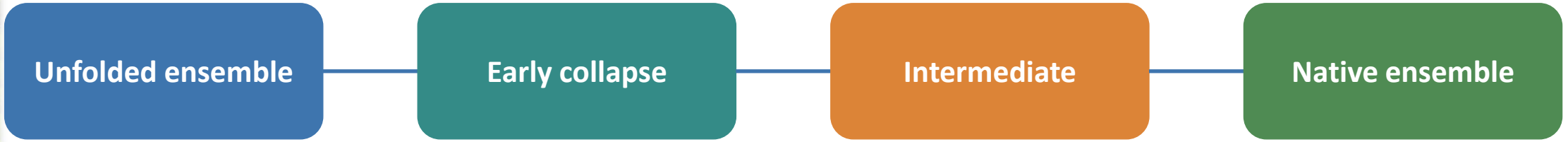
Longer than the age of universe!

In reality…..



- Many proteins fold in seconds, sometimes need help of chaperones.
- Follow directed folding pathway
- Secondary structure fold first
- Hydrophobic collapse
  - Molten globule – much of the correct secondary structure, no tertiary structure.
- Then tertiary structure starts to fold via subomains
- Rearrangements for internal side chains via van der vals interaction.
  - Water expelled from the core
- Energy funnel is a good model.

Folding is often a pathway—not a single jump



Folding kinetics = movement through the landscape

What drives a protein to fold?

Folding reflects a balance of many weak interactions.

Hydrophobic effect

Nonpolar side chains tend to become buried

Hydrogen bonds

Stabilize helices, sheets and other contacts

Electrostatics

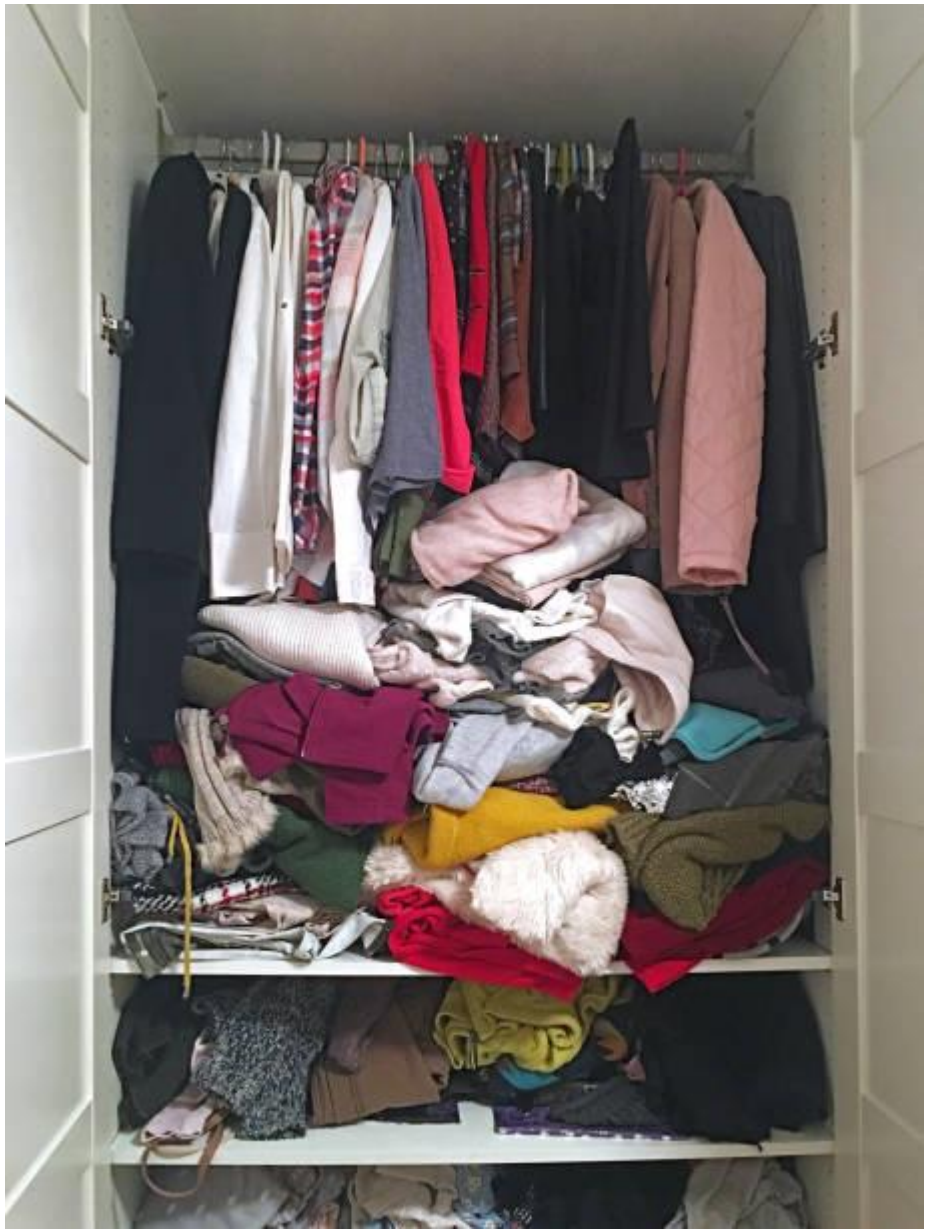
Charged groups attract or repel

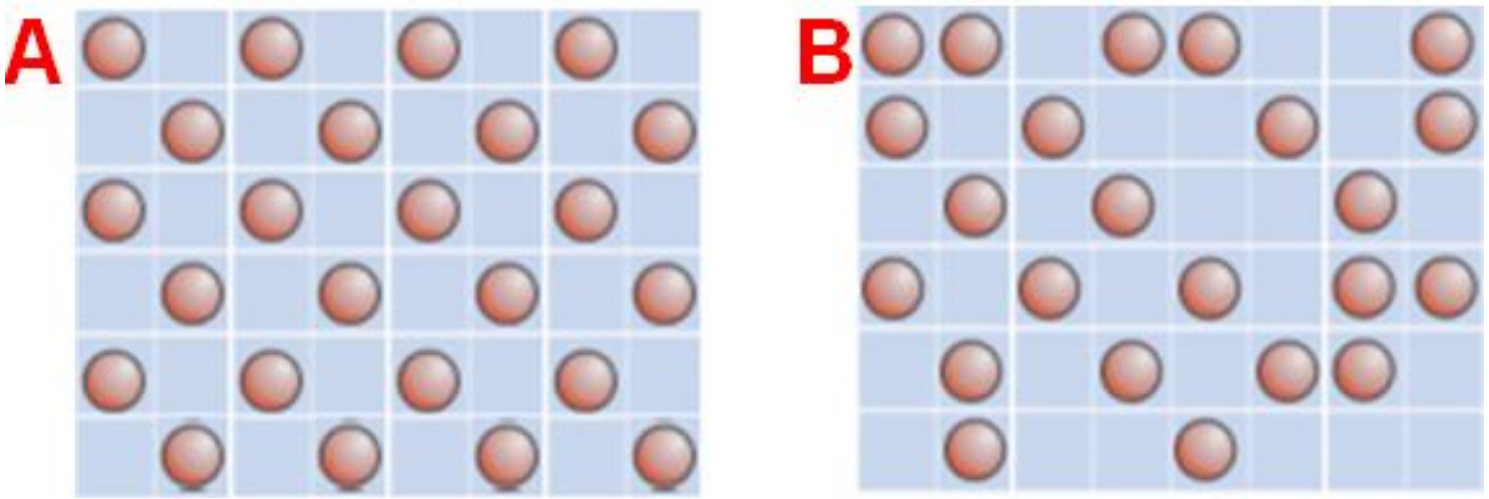
van der Waals

Close packing provides many small contacts

Net result: a lower-free-energy ensemble is favored

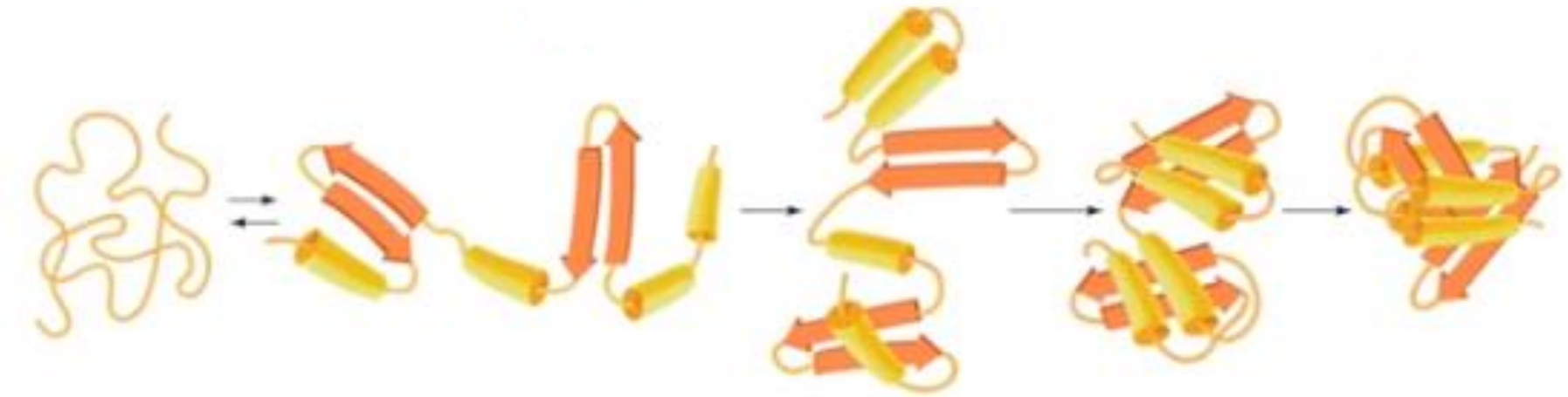






24 molecules, 48 lattices

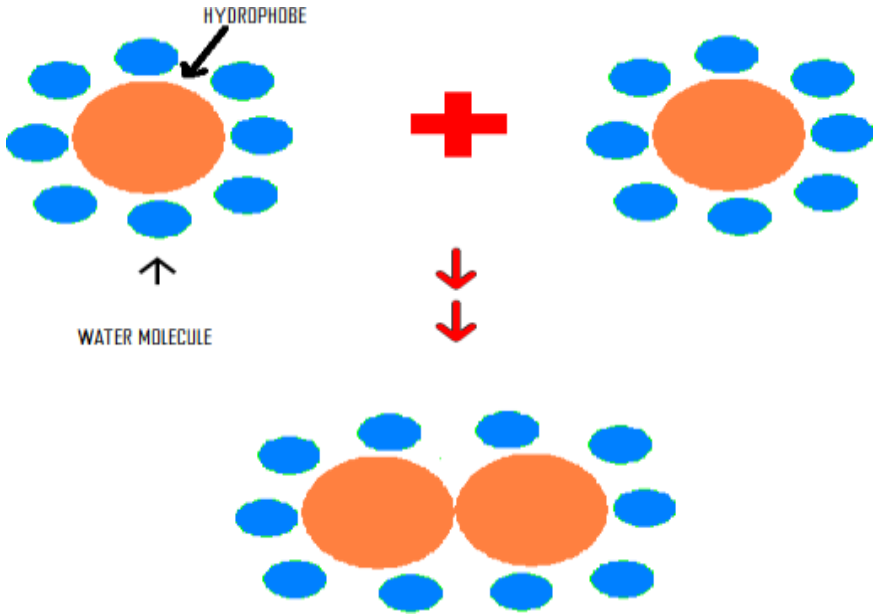
$$S = k_B ln W$$



$$S = k_B ln W$$

Hydrophobic interactions

The non-polar substance like fat molecules tend to clump up together rather than distributing itself in a water medium, because this allow the fat molecules to have minimal contact with water.

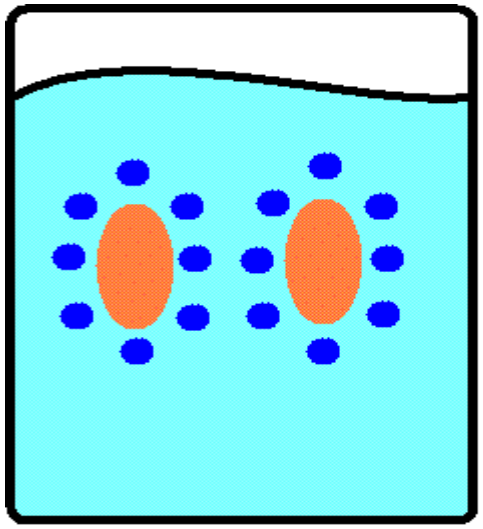


Thermodynamics of Hydrophobic interactions

- When a hydrophobe is dropped in an aqueous medium hydrogen bonds between water molecules will be broken to make space for the hydrophobe.
- This considered an endothermic reaction, because when bonds are broken heat is put into the system.
- Water molecules that are distorted by the pressure of the hydrophobe will make a new Hydrogen bonds and form an ice like cage structure called a ***clathrate cage*** around the hydrophobe.

According to equation $\Delta G = \Delta H - T\Delta S$

If ΔH is small positive value and ΔS is large positive value then ΔG will be negative hence hydrophobic interaction is spontaneous.

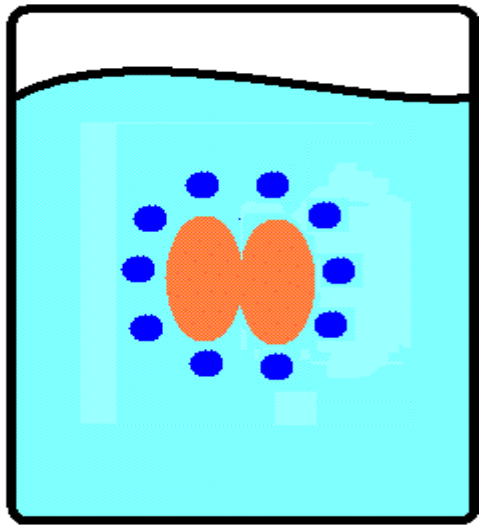


No Interactions

ΔH = Negative

ΔS = Negative

ΔG = Positive



Hydrophobic Interaction

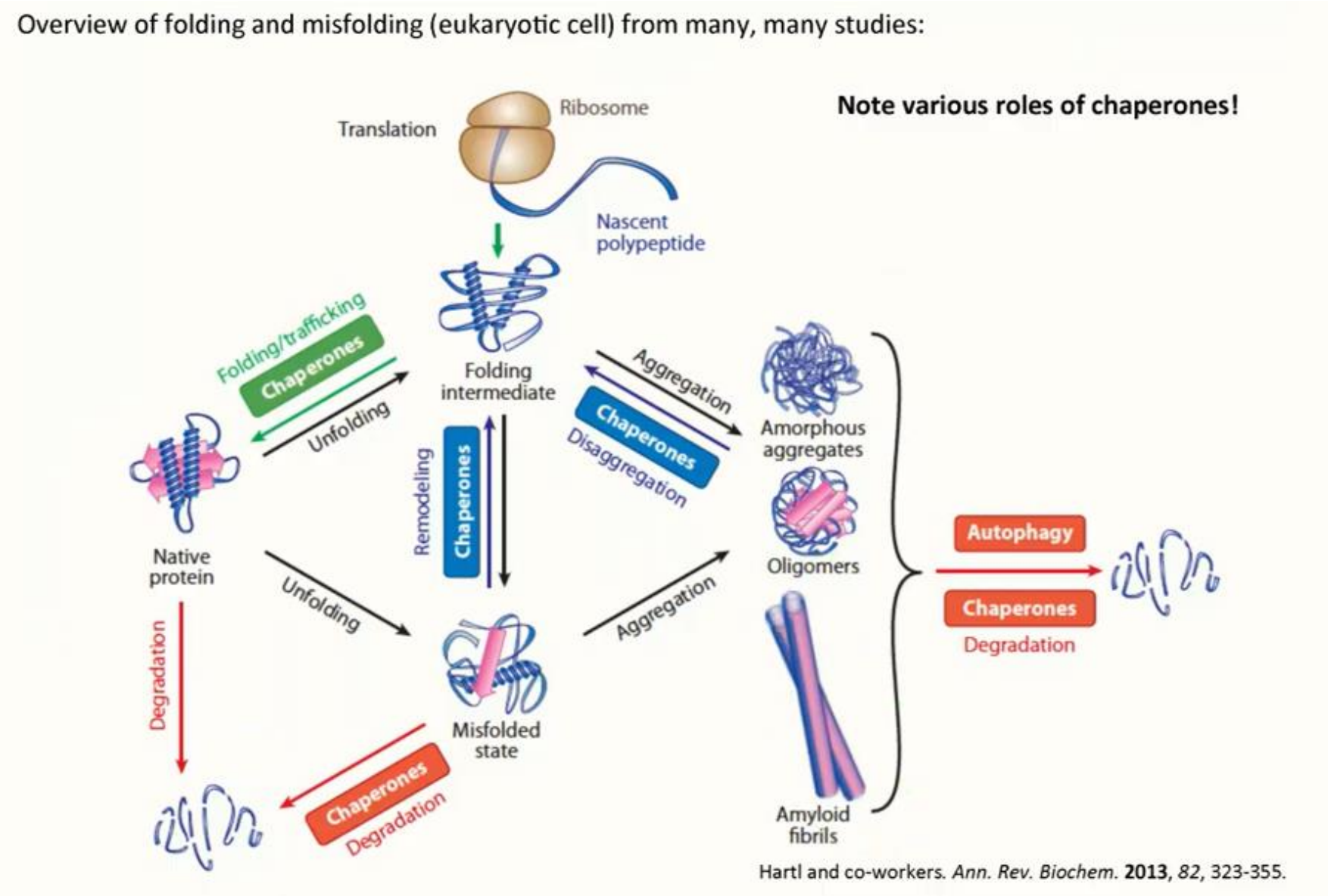
ΔH = Positive

ΔS = Positive

ΔG = Negative

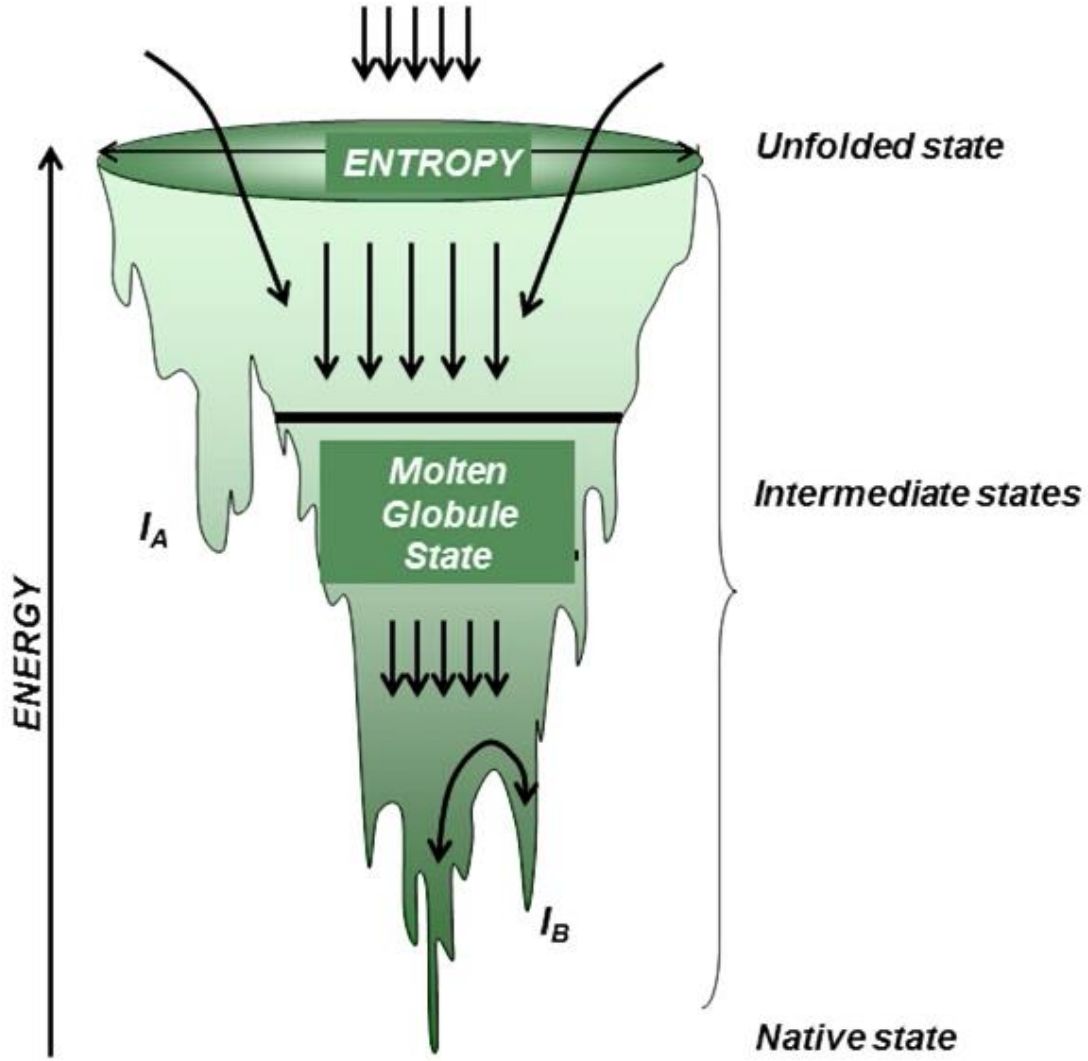
Protein folding and misfolding – a journey of various paths!

Overview of folding and misfolding (eukaryotic cell) from many, many studies:



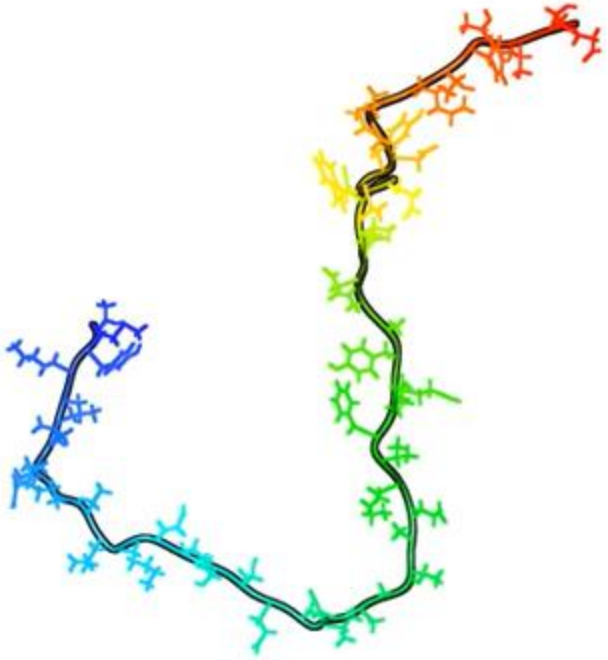
Hartl and co-workers. *Ann. Rev. Biochem.* **2013**, *82*, 323-355.

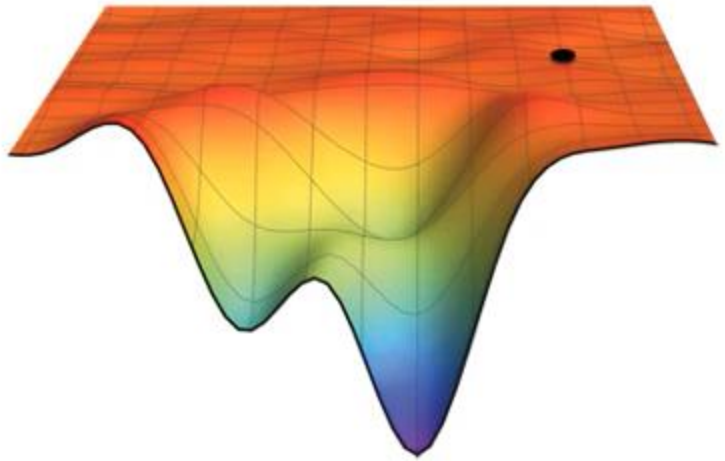
Protein folding: energy landscape



1. Fast – (on a ms timescales for single domains). Unfolded proteins "roll downhill" towards smaller populations of conformations.
2. Highly cooperative – intermediates are rarely observed
3. Heterogeneity of the starting points – each unfolded molecule has different conformation and different path downhill the folding funnel
4. In many cases is coupled to translation

Protein folding: energy landscape





https://www.youtube.com/watch?v=YANAso8Jxrk

Protein Folding: Key Takeaways

From a linear polypeptide to a functional three-dimensional structure

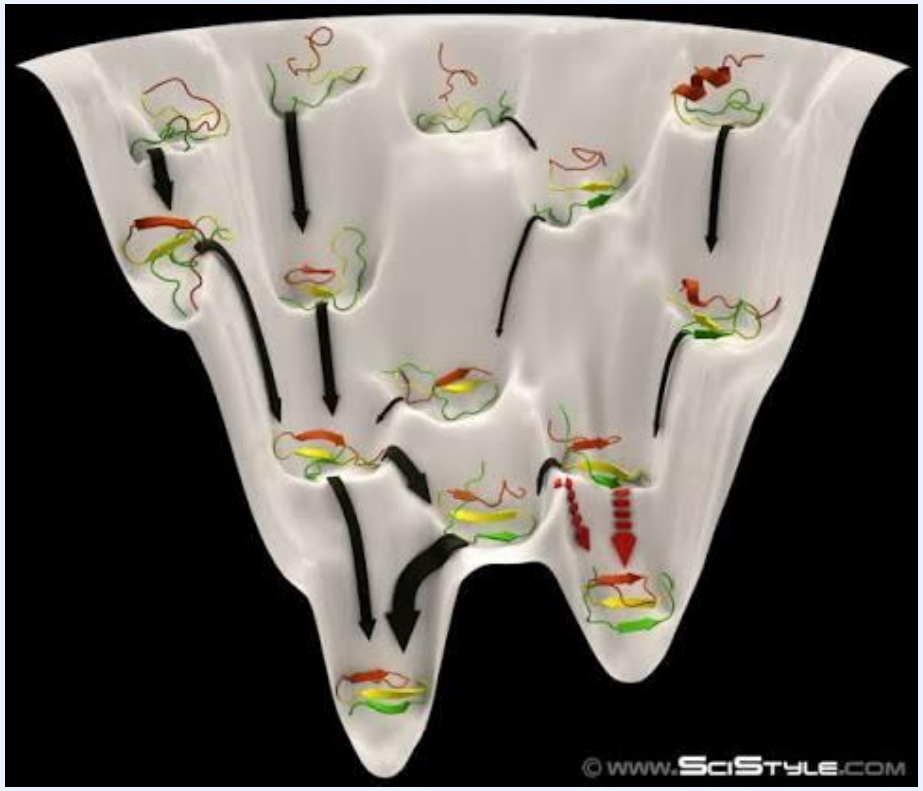
1. **Sequence encodes structure**
   A newly synthesized polypeptide begins as a flexible amino-acid chain and adopts a specific 3D conformation.

2. **Folding can be remarkably rapid**
   Many small proteins reach the native state in milliseconds to seconds rather than sampling every possible conformation.

3. **Some proteins need assistance**
   Molecular chaperones reduce aggregation and help difficult proteins fold, especially in the crowded cellular environment.

4. **Folding is a thermodynamic balance**
   The native state is favored when the overall Gibbs free-energy change is negative: $\Delta G = \Delta H - T\Delta S$.

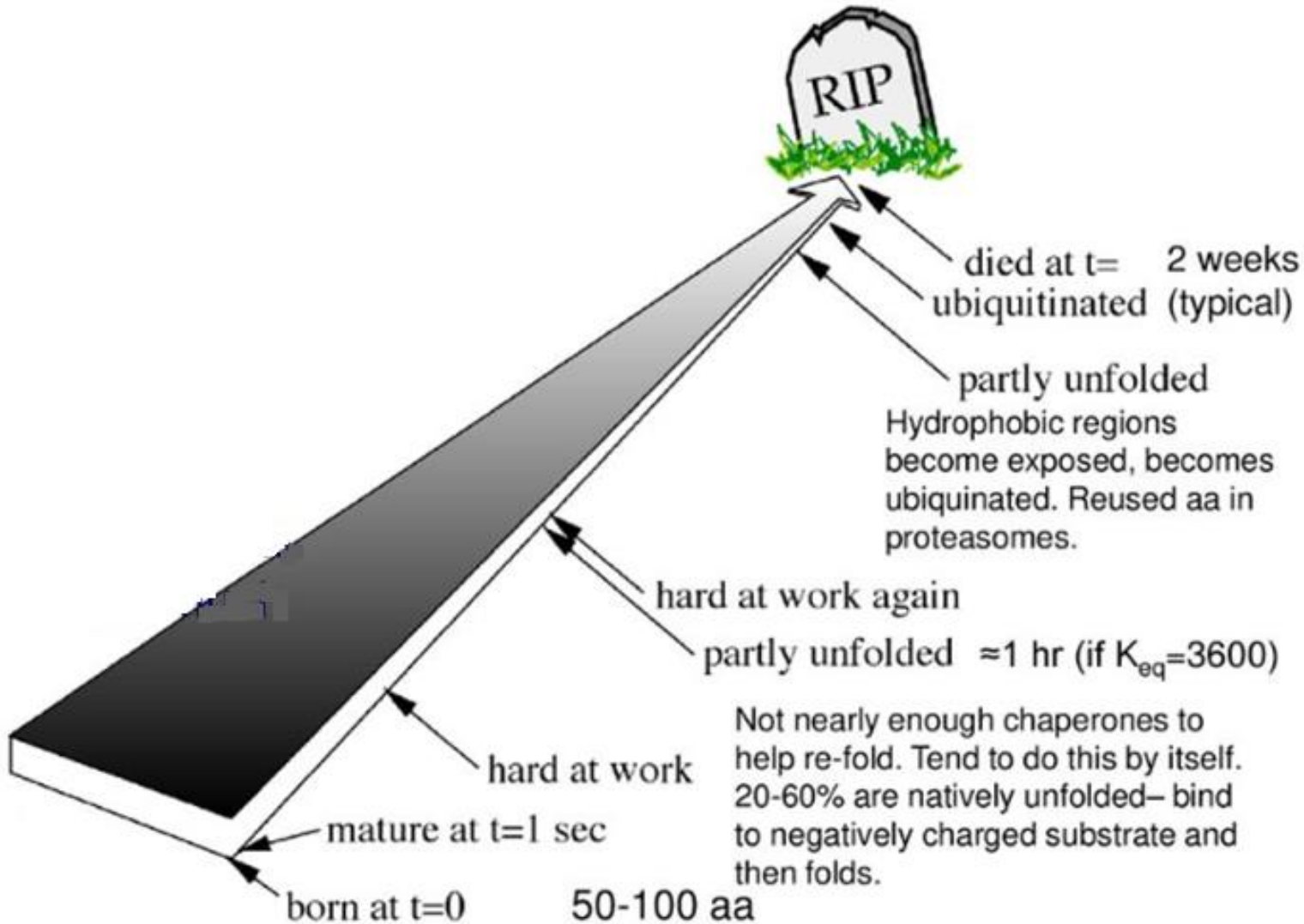
5. **The energy landscape is funnel-shaped**
   Many routes can converge on the low-free-energy native structure; kinetic barriers may create intermediates or traps.



Many conformations → lower free energy → native state

Core idea: folding is biased toward the native state, but the route can include intermediates and chaperone-assisted steps.

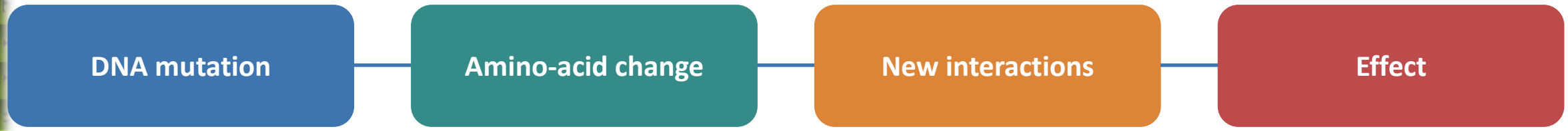
Proteins: A short, hard life.



What Are Mutations?

- Changes in the nucleotide sequence of DNA
- May occur in somatic cells (aren't passed to offspring)
- May occur in gametes (eggs & sperm) and be passed to offspring
- May be due to copying errors, chemicals, viruses, etc.

Mutations: changing the sequence changes the landscape



A mutation may alter stability, folding kinetics, binding, catalysis, localization—or have little measurable effect.

Conservative vs non-conservative substitutions

Conservative: similar chemical properties

Leu → Ile: both hydrophobic

Lys → Arg: both positively charged

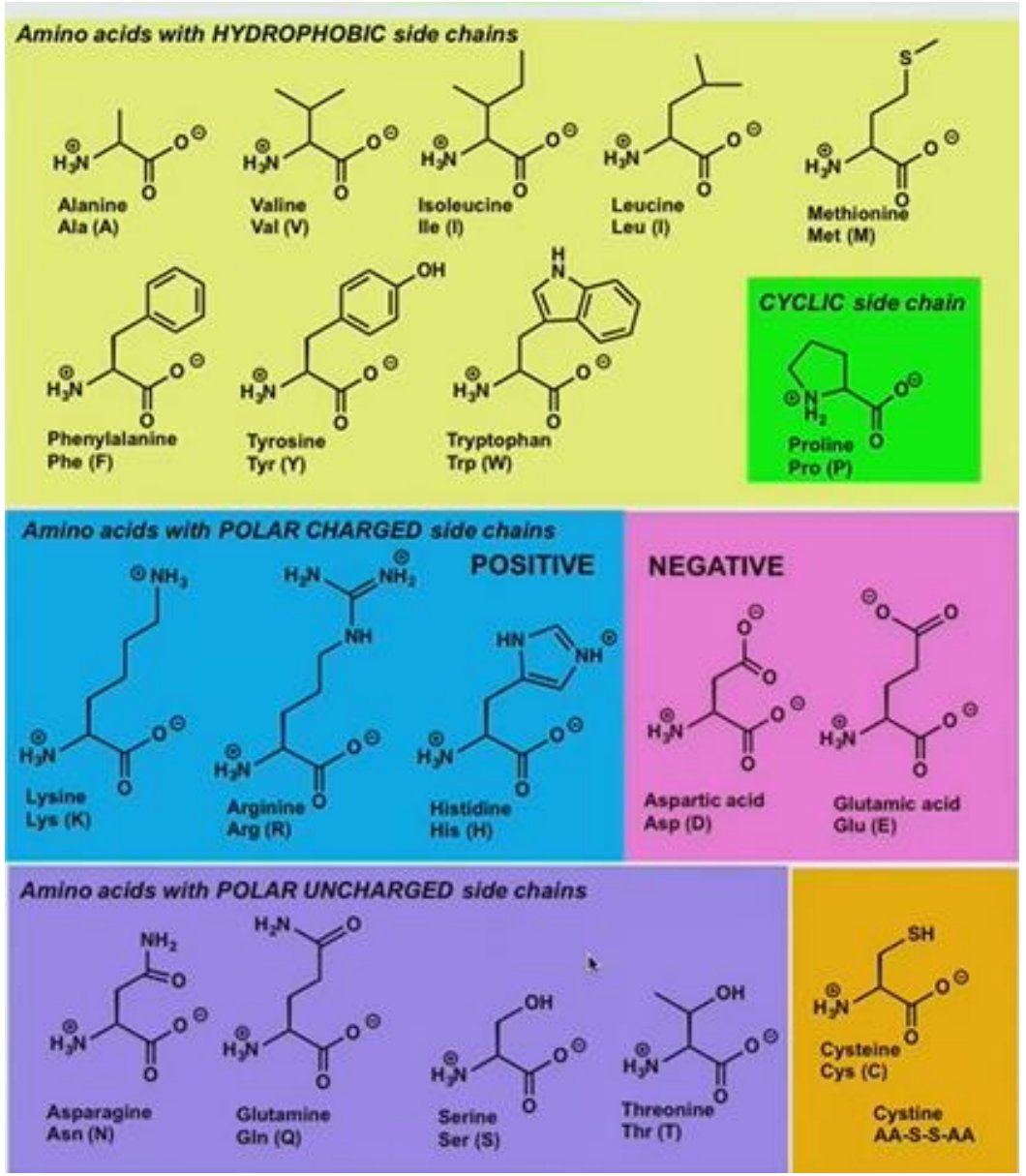
Often smaller structural perturbation

Non-conservative: different properties

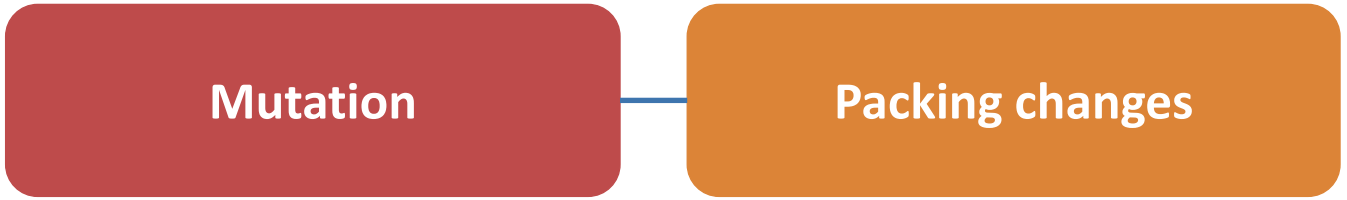
Val → Asp: hydrophobic → negatively charged

Gly → Trp: very small → bulky aromatic

May introduce strong energetic or packing changes



Mutation example: the hydrophobic core

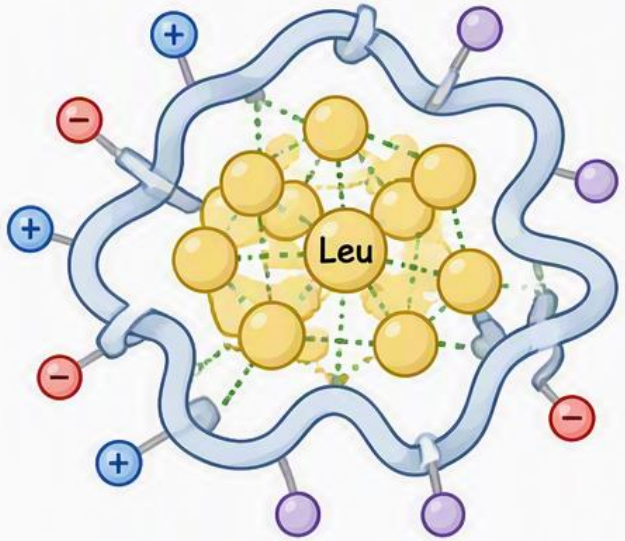


Possible outcomes: cavity • steric clash • altered stability

Effect of a Mutation in the Hydrophobic Core of a Protein

Hydrophobic cores are tightly packed with nonpolar side chains.
A mutation that introduces a polar or charged residue can destabilize the protein.

WILD TYPE (Stable)



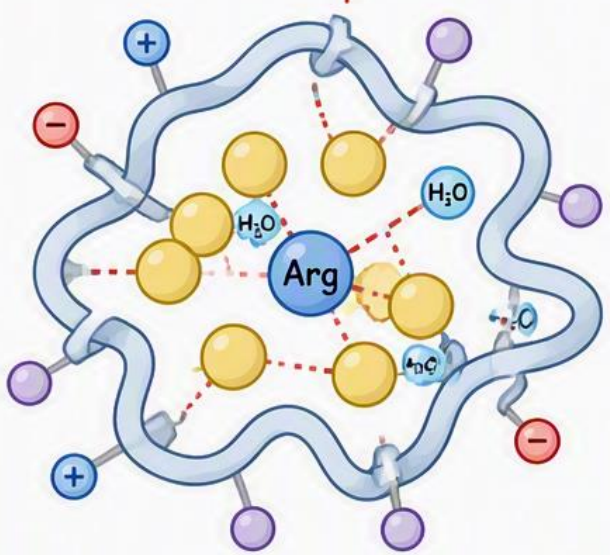
- ☑ Nonpolar residues packed in the core
- ☑ Favorable hydrophobic interactions
- ☑ Proper folding and stability

Protein is correctly folded and functional

MUTATION

e.g., Leu → Arg

MUTANT (Destabilized)



- ☒ Polar/charged residue in hydrophobic core
- ☒ Unfavorable interactions and water penalty
- ☒ Destabilized structure, loss of function

Protein is destabilized and may lose function

Legend

- Hydrophobic residue (nonpolar)
- Positively charged (residue)
- Negatively charged (residue)
- Polar residue
- Favorable hydrophobic / van der Waals contacts
- Unfavorable interactions
- Water

Key takeaway

Mutations in the hydrophobic core that introduce polar or charged residues can disrupt packing, allow water to enter, and destabilize the protein. Even a single substitution can significantly reduce stability and function.

Mutation example: active site vs structural site

Mutation far from active site

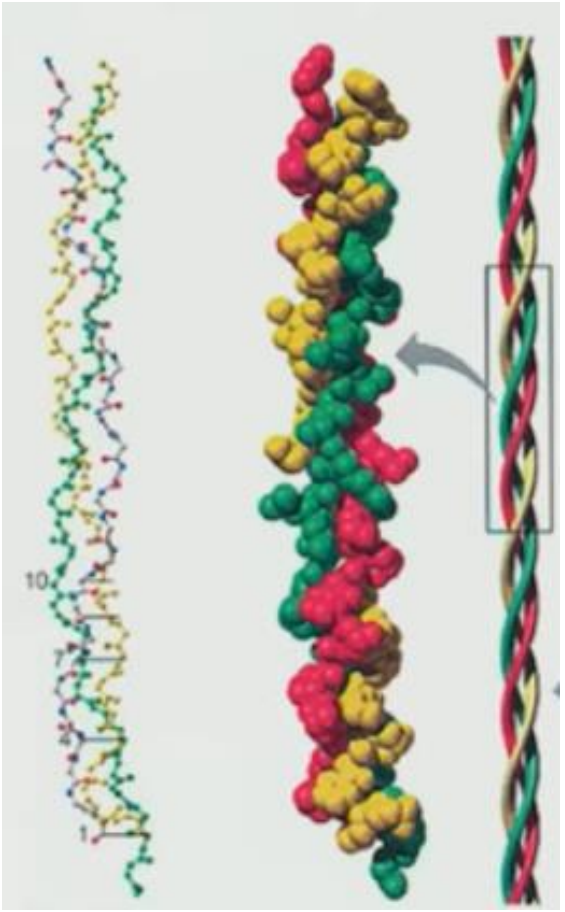
Structure may remain similar
but stability can change

Mutation at active site

Small geometric / chemical change
can strongly alter catalysis

Fibrous protein example - Collagen

- Collagen is a structural protein found in bones, tendons, intestinal wall, cornea, heart valve etc.
- Single amino acid change in Primary sequence can destabilize tertiary and quaternary structure and have a major effect on function
- Primary sequence mutations are common cause of inherited diseases

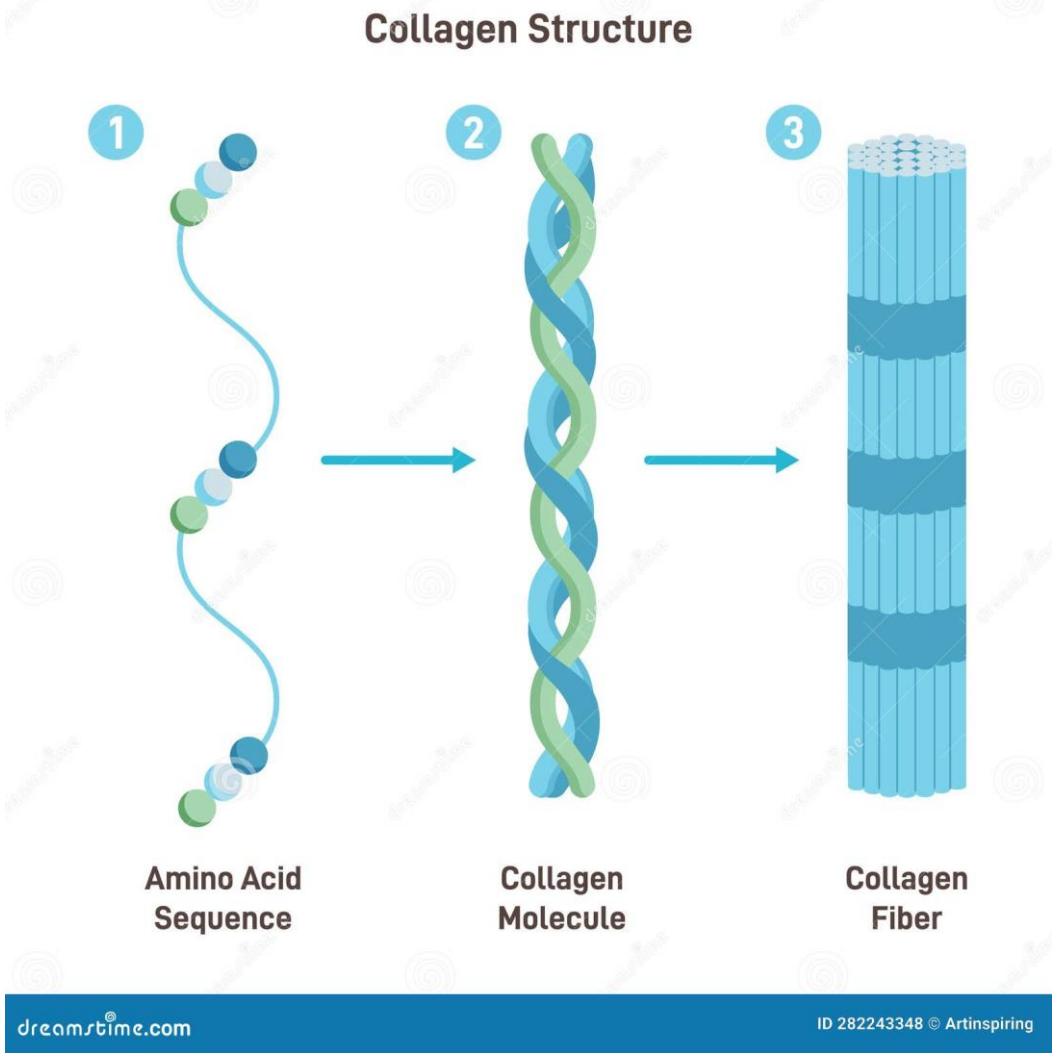


Fibrous protein example - Collagen

- Collagen is a structural protein found in bones, tendons, intestinal wall, cornea, heart valve etc.
- Single amino acid change in Primary sequence can destabilize tertiary and quaternary structure and have a major effect on function
- Primary sequence mutations are common cause of inherited diseases



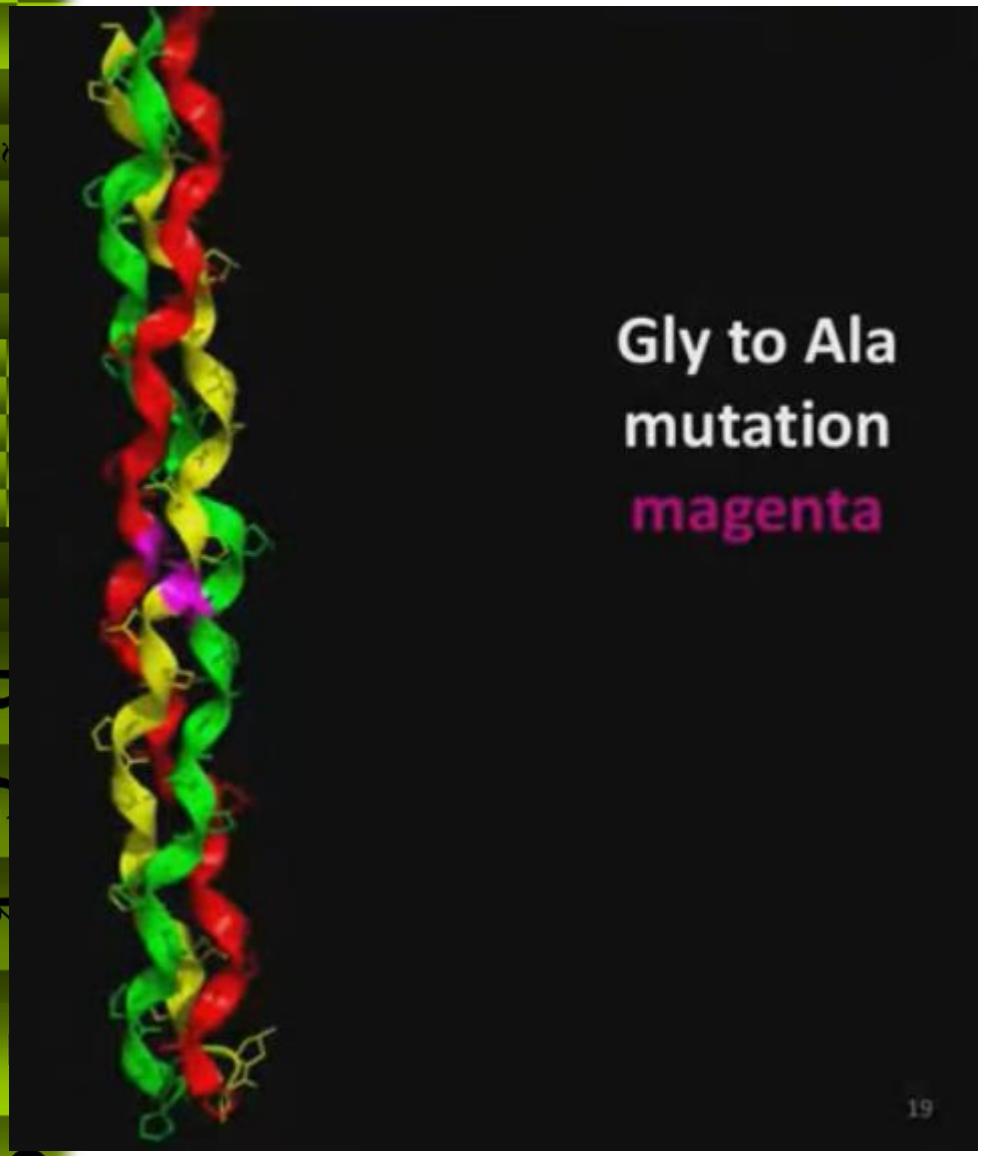
Packaging of Collagen

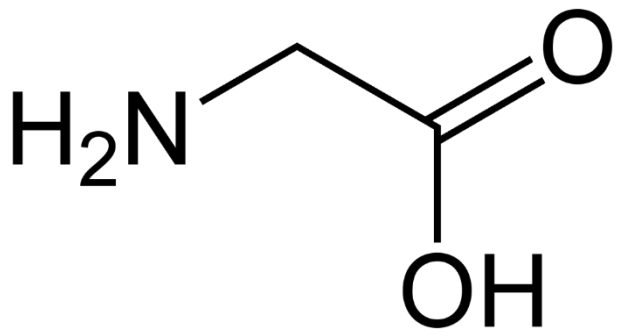


Seq: Gly-X-Y

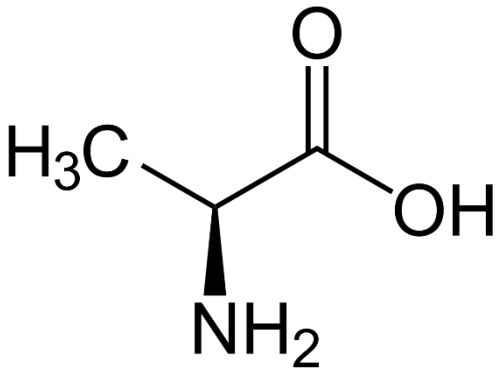
X – Proline
Y – Hydroxyproline

Fibrous protein example - Collagen





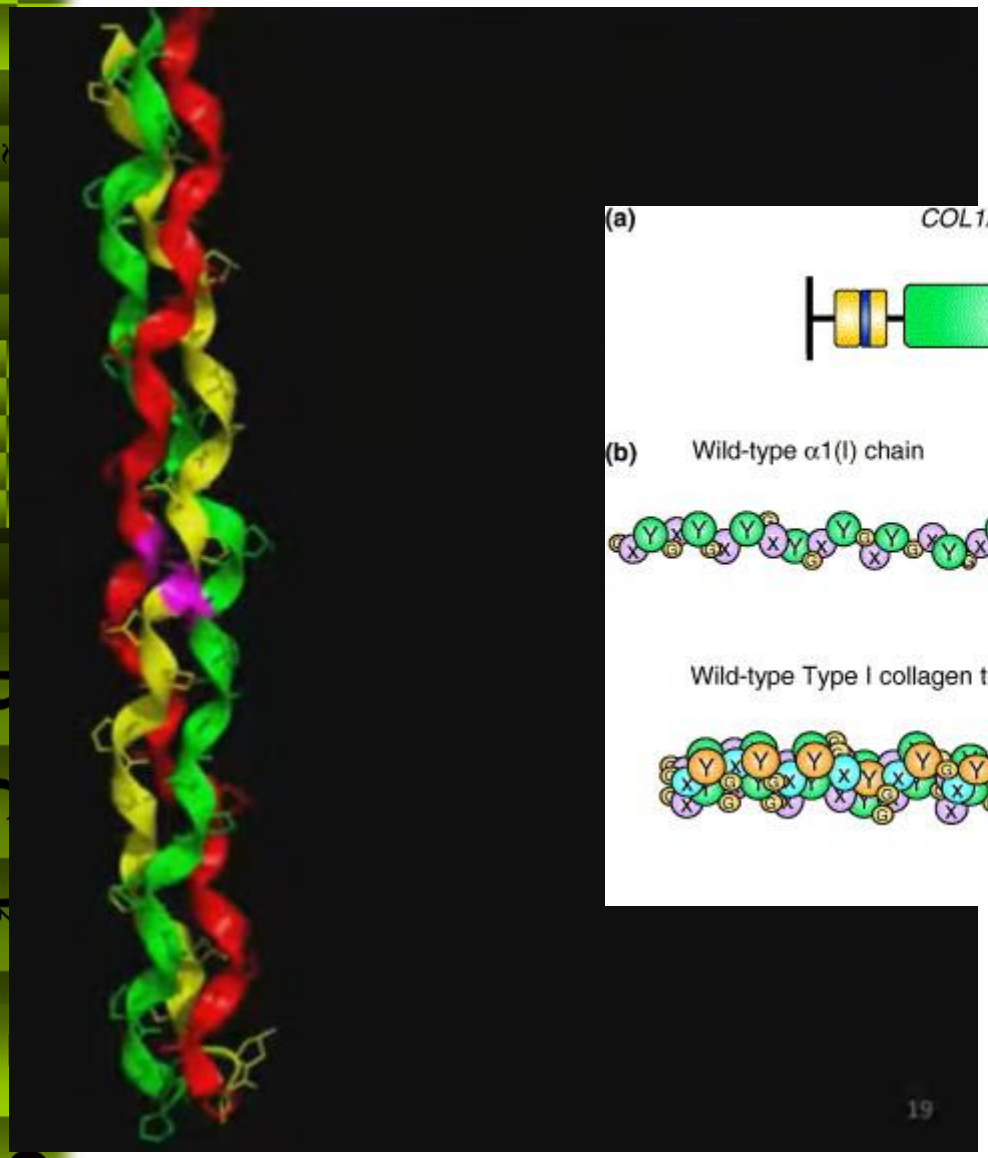
Glycine

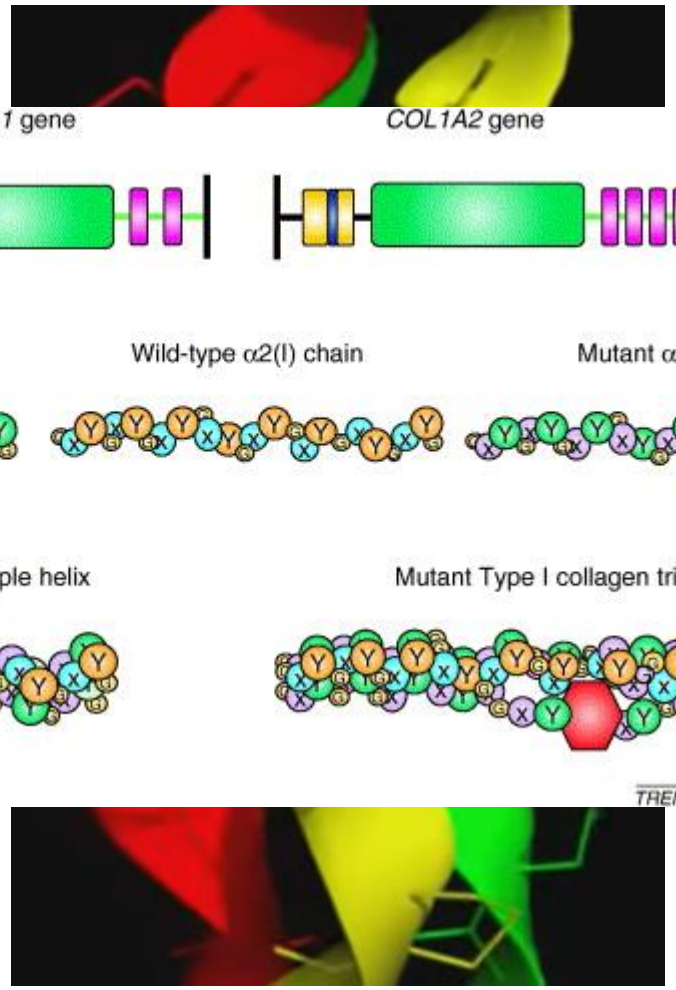


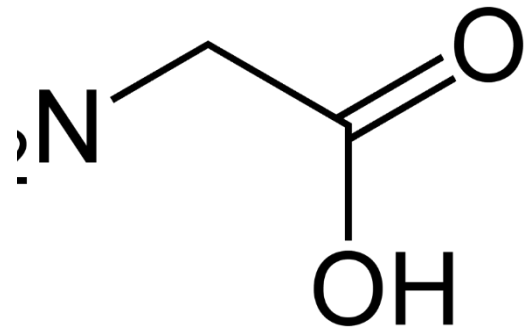
Alanine

https://www.cell.com/trends/molecular-medicine/fulltext/S1471-4914(05)00097-3

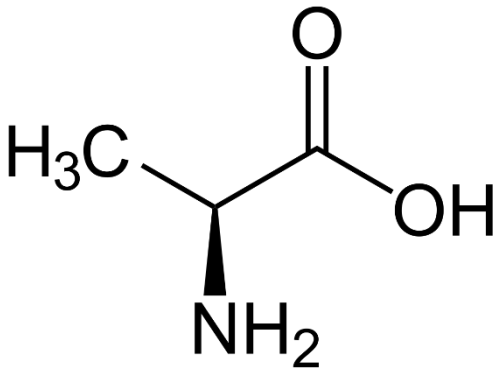
Fibrous protein example - Collagen







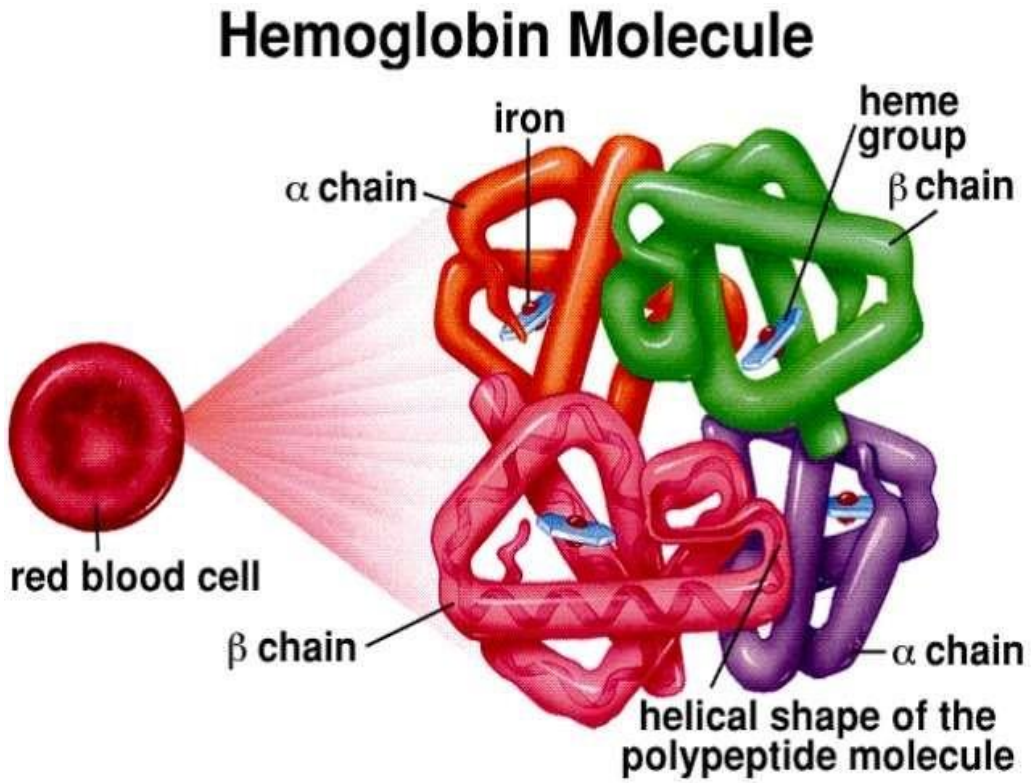
Glycine



Alanine

Globular protein example - Haemoglobin proteins

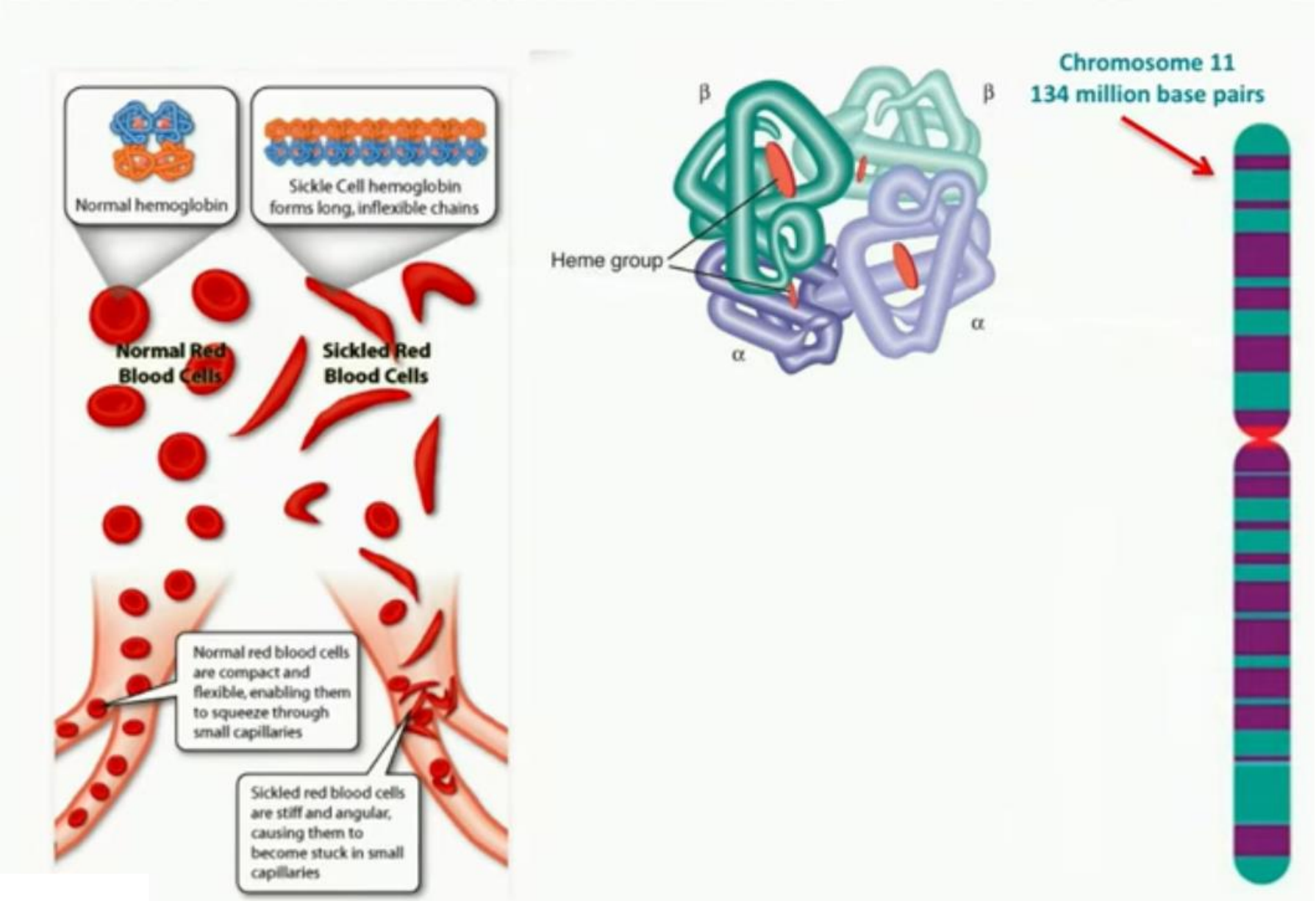




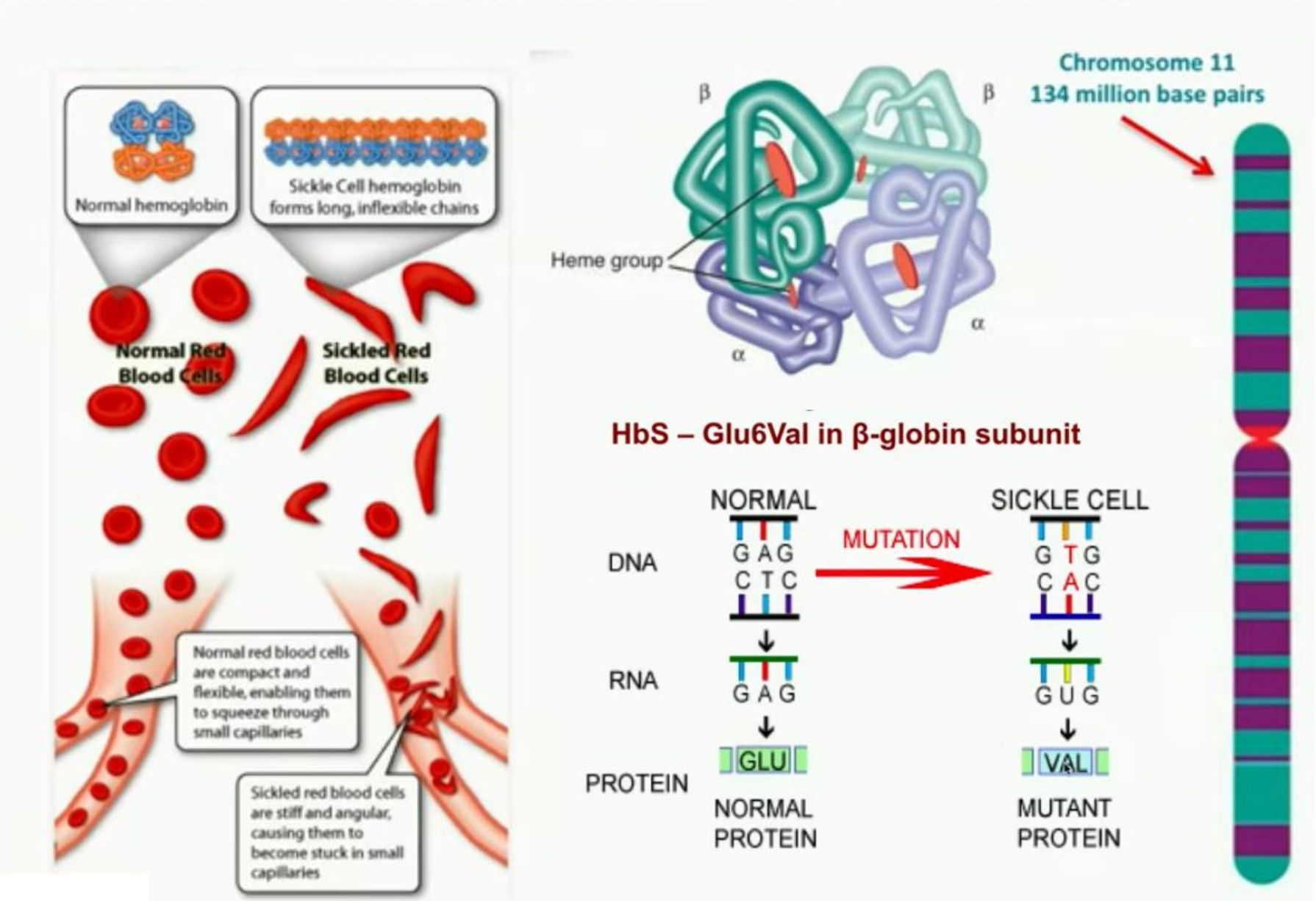
S Raj et al., Cell biochemistry and biophysics; 65: (2013)

S Raj et al., Biomed. Opt. Express; 3: (2012)

Globular protein example - Haemoglobin proteins



Globular protein example - Haemoglobin proteins



Globular protein example - Haemoglobin proteins

